

通用音频大模型

从“通用感知”到“听说一体”

于文一

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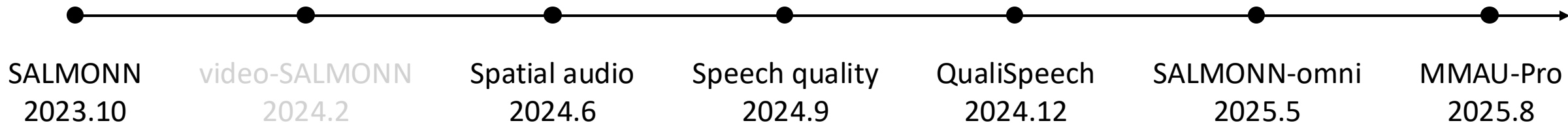
通用音频大模型

通用感知

- 语音、音频、音乐
- 低端听觉
- 空间音频
- 视听感知

听说一体

- 无codec注入的全双工对话框架
- SALMONN-omni



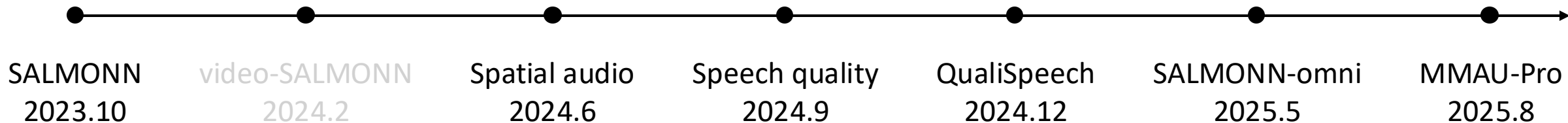
通用音频大模型

通用感知

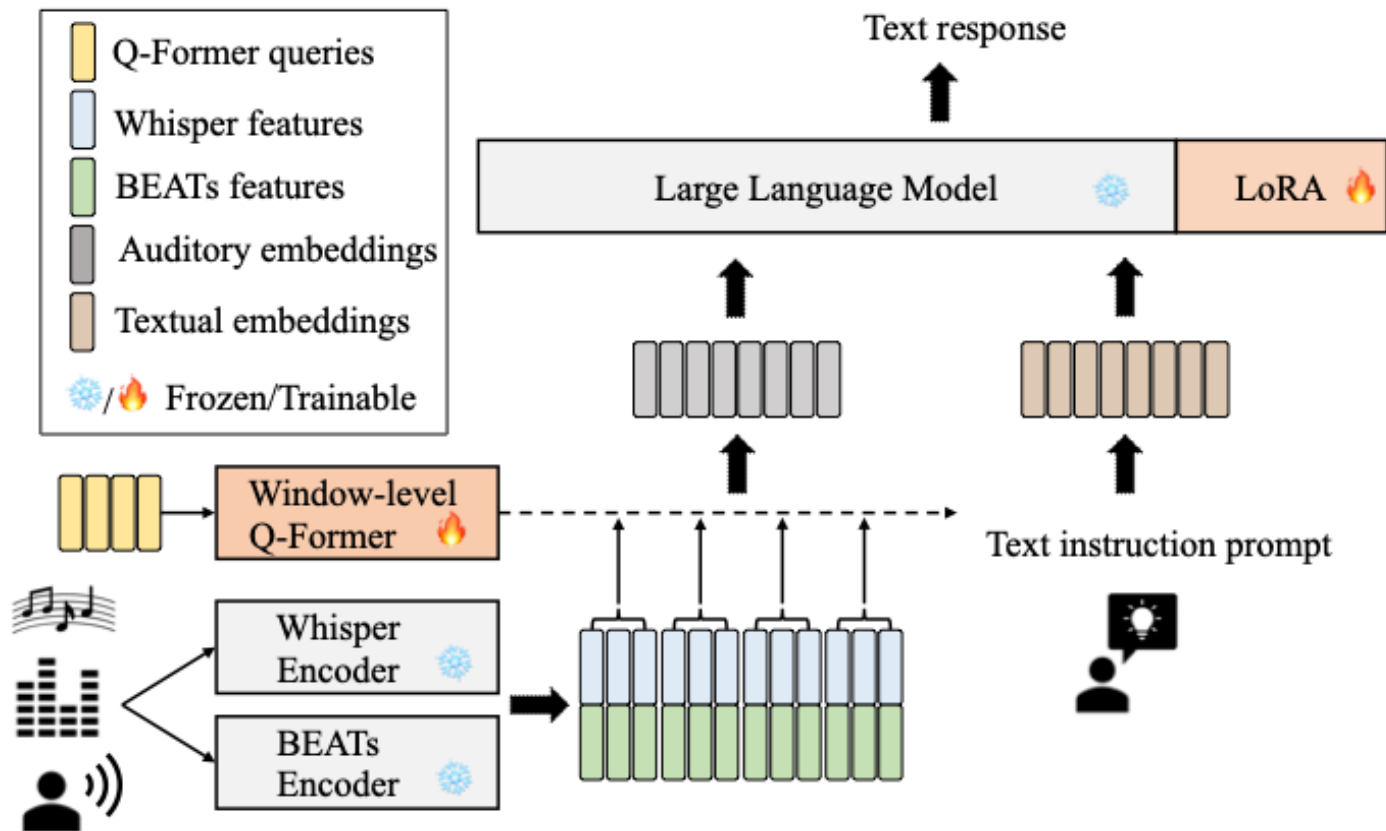
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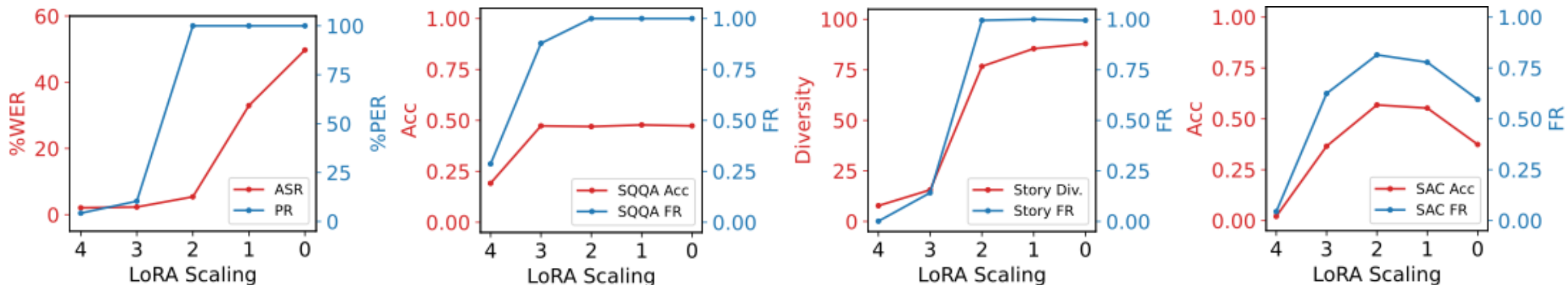


语音、音频、音乐



Tang et al., SALMONN: Towards generic hearing abilities for large language models, ICLR 2024.

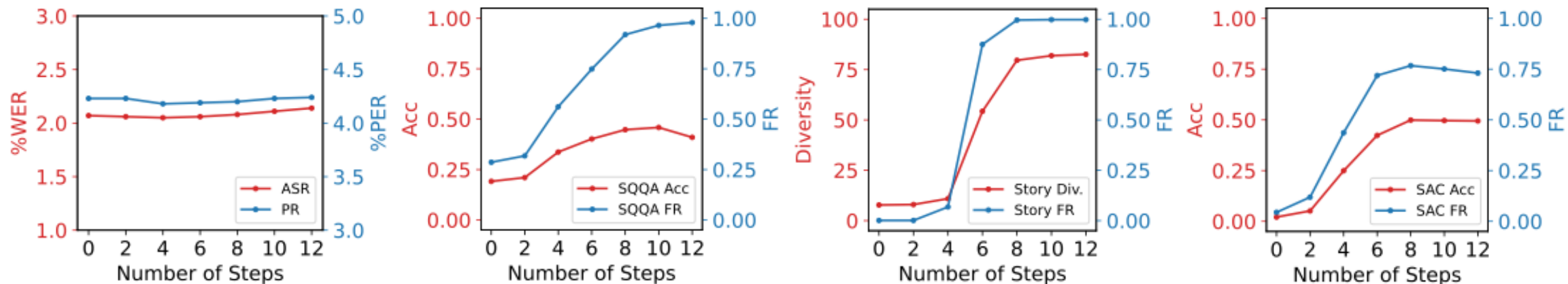
LLM能力的保持（知识、指令遵循能力）



降低LoRA的权重是一种削弱模态过拟合现象的方法

<https://www.youtube.com/watch?v=kGShfZNUQ3g>

LLM能力的保持（知识、指令遵循能力）



使用一些音频文本耦合不太紧密的任务也可以一定程度缓解该问题

LLM能力的保持（知识、指令遵循能力）

预训练任务	test-clean	test-other	en2zh	en2de
ASR	3.96	6.37	34.80	x
ASR+NU	2.89	5.15	38.44	x
NU	2.87	5.69	38.64	22.54
NU (GPT)	2.68	5.54	38.34	x

训练任务、训练方式影响模态间的耦合

预训练任务不同，指令微调都使用ASR、en2zh、QA

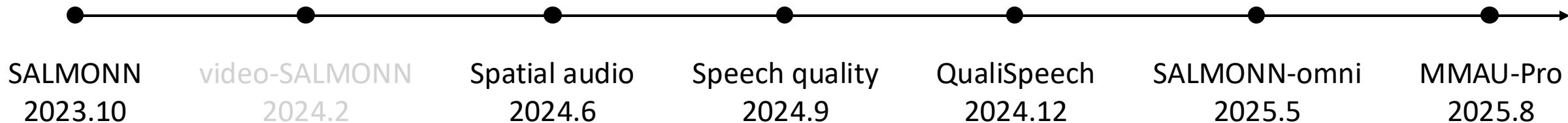
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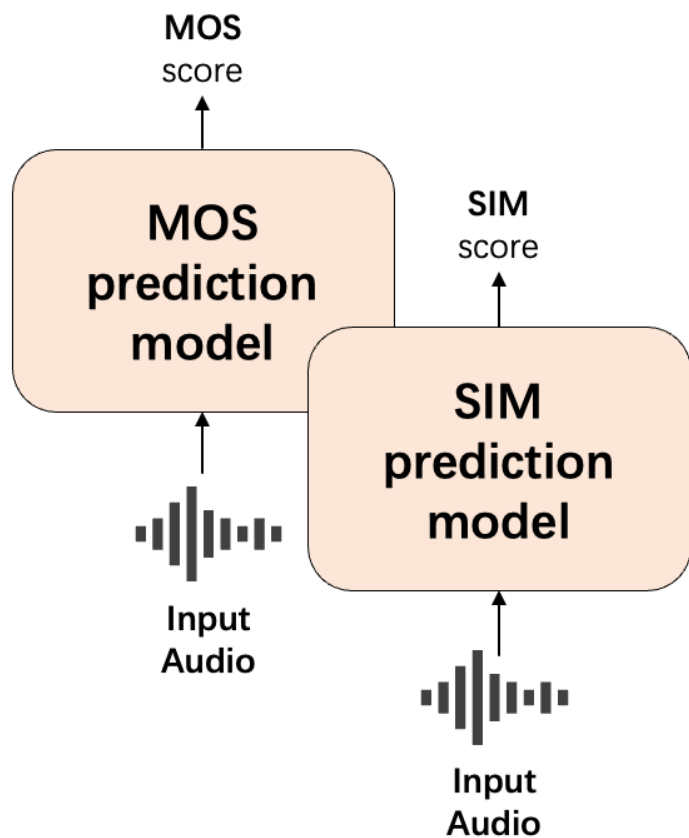
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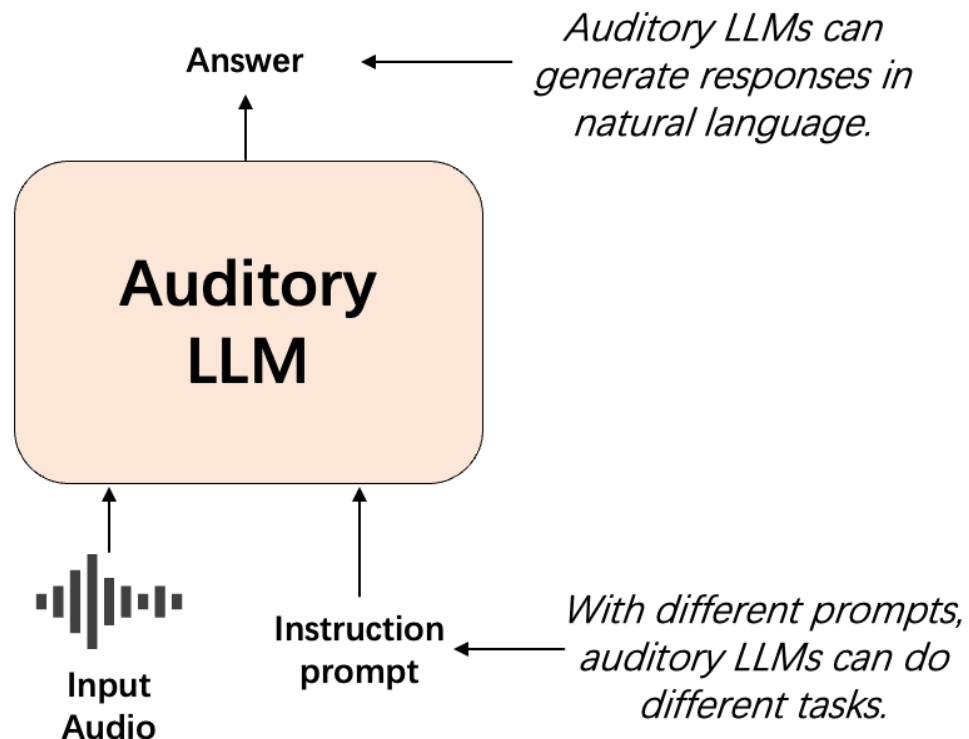
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语音质量评估

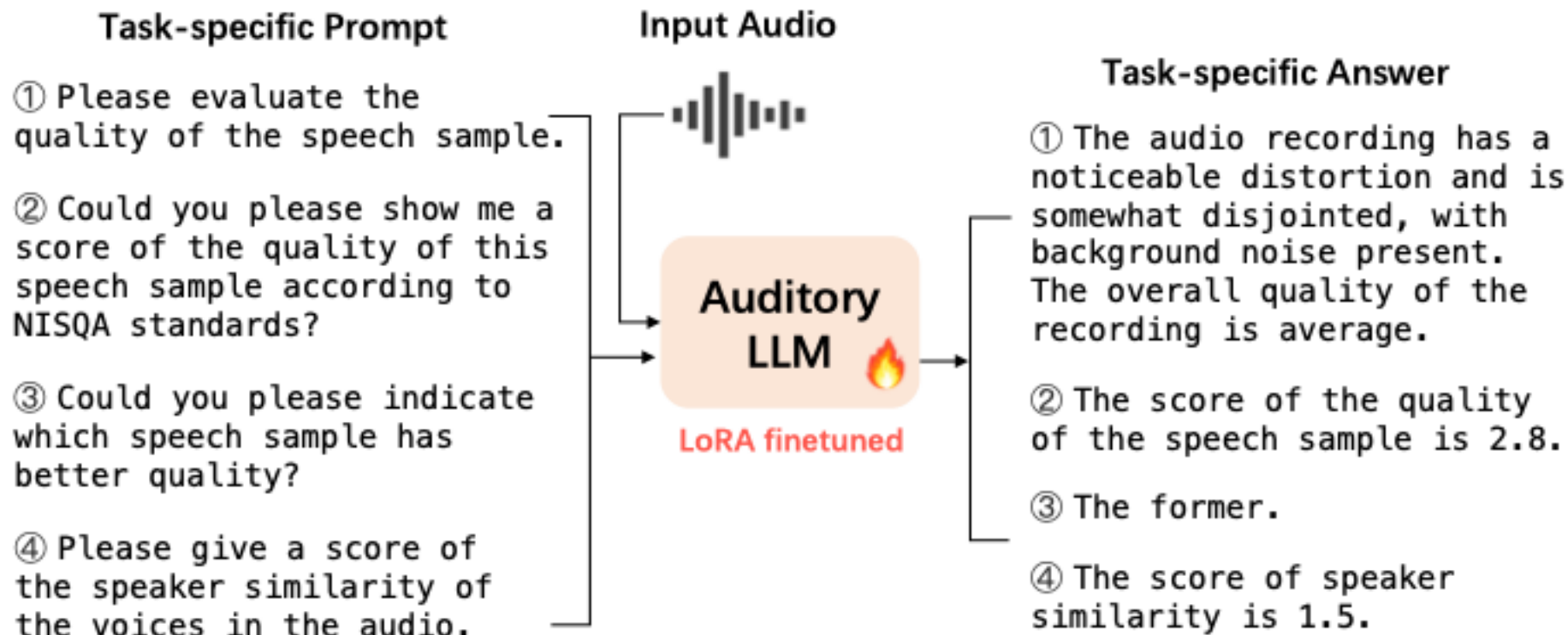


② Quality assessment in natural language



① Multi-task speech quality assessment

语音质量评估



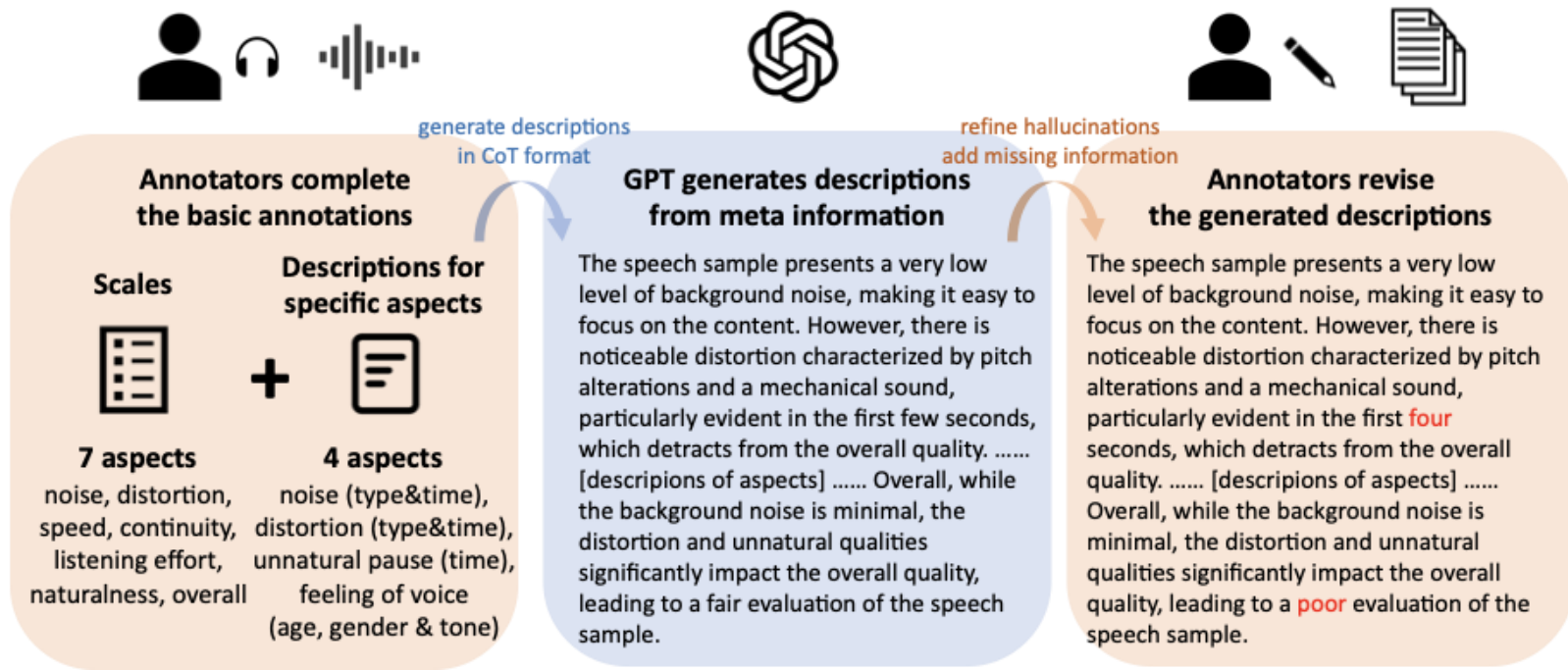
Wang et al., Enabling Auditory Large Language Models for Automatic Speech Quality Evaluation, ICASSP 2025.

语音质量评估

Model	NISQA			BVCC						SOMOS					
	utterance-level			utterance-level			system-level			utterance-level			system-level		
	LCC	SRCC	MSE	LCC	SRCC	MSE	LCC	SRCC	MSE	LCC	SRCC	MSE	LCC	SRCC	MSE
Single-task SOTA	0.894	0.887	0.286	0.899	0.896	0.165	0.939	0.936	0.090	0.687	0.681	0.203	0.911	0.917	0.052
WavLM-Base+	0.739	0.724	0.536	0.769	0.765	0.350	0.805	0.799	0.251	0.349	0.318	0.288	0.544	0.539	0.081
WavLM-Large	0.850	0.845	0.467	0.813	0.805	0.377	0.884	0.879	0.189	0.623	0.613	0.257	0.880	0.880	0.044
SALMONN (vic1.5)	0.861	0.859	0.347	0.826	0.833	0.282	0.884	0.884	0.152	0.644	0.636	0.196	0.894	0.891	0.034
SALMONN (vic1.0)	0.829	0.831	0.453	0.819	0.827	0.296	0.860	0.864	0.181	0.631	0.626	0.200	0.902	0.904	0.030
Qwen-Audio	0.771	0.784	0.664	0.680	0.685	0.451	0.790	0.803	0.375	0.569	0.553	0.225	0.834	0.858	0.036
Qwen2-Audio	0.768	0.780	0.643	0.681	0.678	0.493	0.800	0.797	0.247	0.583	0.572	0.216	0.850	0.873	0.040

微调后的ALLM在多任务、多数据集上可以实现较好的语音质量评估

低端听觉感知



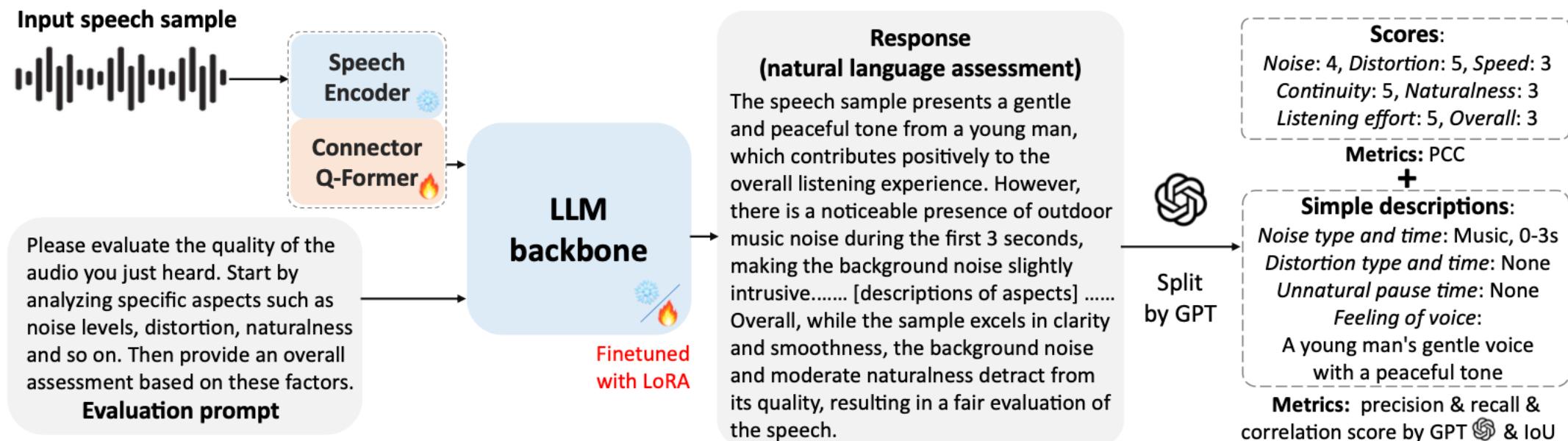
Wang et al., QualiSpeech: A Speech Quality Assessment Dataset with Natural Language Reasoning and Descriptions, ACL 2025.

低端听觉感知

Model	Aspect of low-level speech perception						Overall
	Noise	Distortion	Speed	Continuity	Effort	Naturalness	
SALMONN-7B	0.003	0.013	0.001	nan	nan	0.030	0.084
SALMONN-13B	0.001	0.002	0.025	-0.001	-0.069	0.013	0.100
Qwen-Audio-Chat	0.014	-0.003	0.017	0.145	0.150	0.148	0.250
Qwen2-Audio-7B-Instruct	-0.048	0.056	0.111	0.201	0.035	-0.082	0.112
WavLLM	-0.021	-0.069	0.003	-0.001	-0.007	0.005	0.071

未经训练前，模型几乎无法对这些方面做出有效评估

低端听觉感知



低端听觉感知

Comments	Aspect of low-level speech perception						
	Noise	Distortion	Speed	Continuity	Effort	Naturalness	Overall
revised concise	0.656	0.579	0.212	0.452	0.496	0.568	0.630
concise with num	0.703	0.571	0.178	0.450	0.513	0.535	0.622

模型需要低端听觉感知数据来掌握相关知识
模型的推理能力仍有待提升

Comments	Noise				Distortion				Unnatural pause			Voice	
	Prec	Rec	Corr	IoU	Prec	Rec	Corr	IoU	Prec	Rec	IoU	Corr	GenderAcc
revised concise	0.40	0.50	0.49	0.49	0.80	0.74	0.67	0.78	0.54	0.52	0.34	0.48	0.97
concise with num	0.62	0.54	0.53	0.73	0.76	0.84	0.66	0.77	0.55	0.60	0.34	0.48	0.98
concise	0.34	0.52	0.49	0.41	0.74	0.81	0.65	0.73	0.58	0.57	0.30	0.49	0.96
detailed	0.50	0.27	0.46	0.61	0.76	0.74	0.68	0.73	0.60	0.56	0.33	0.51	0.98

(b) aspects annotated in descriptions

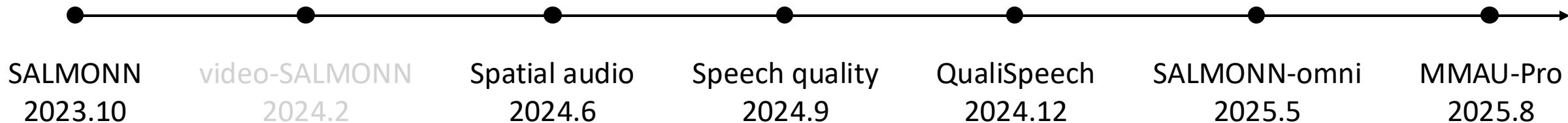
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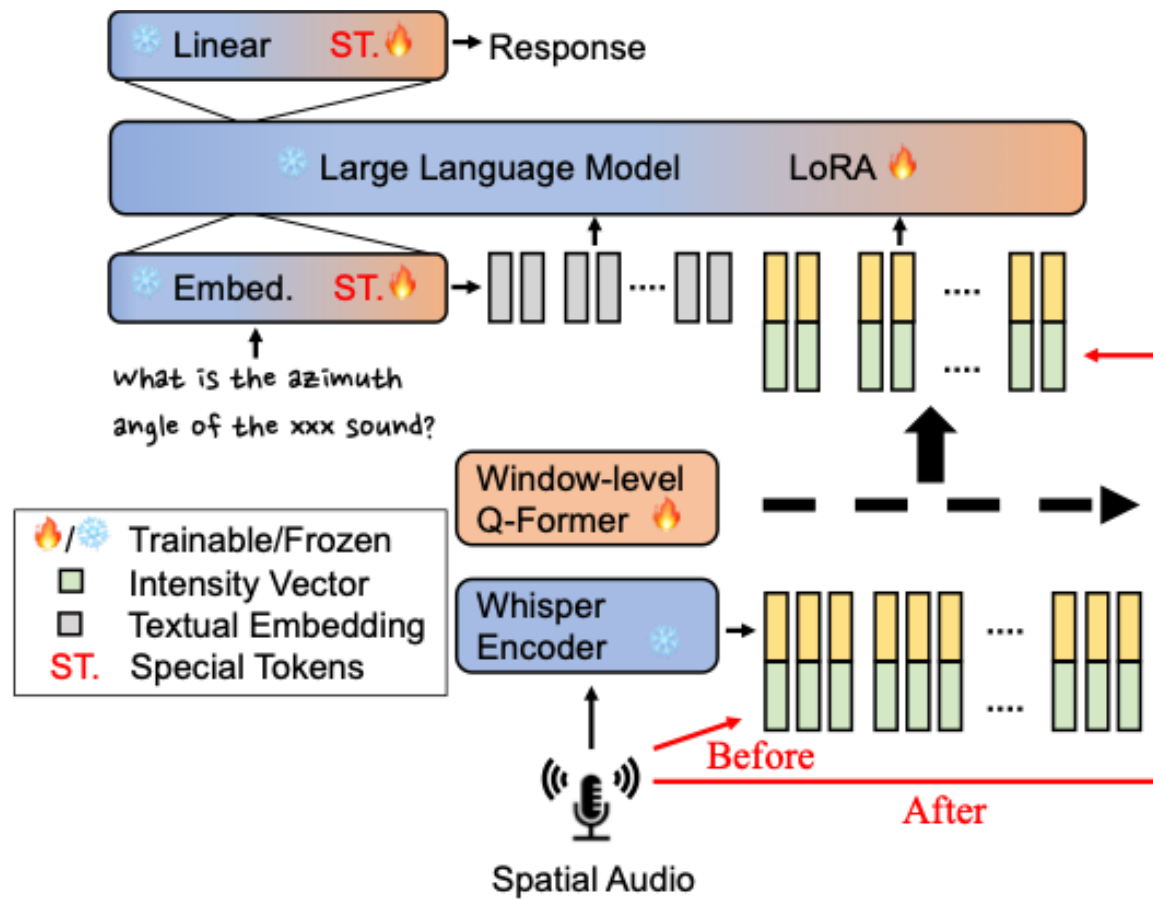
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空间音频



Tang et al., Can Large Language Models Understand Spatial Audio, InterSpeech 2024.

空间音频

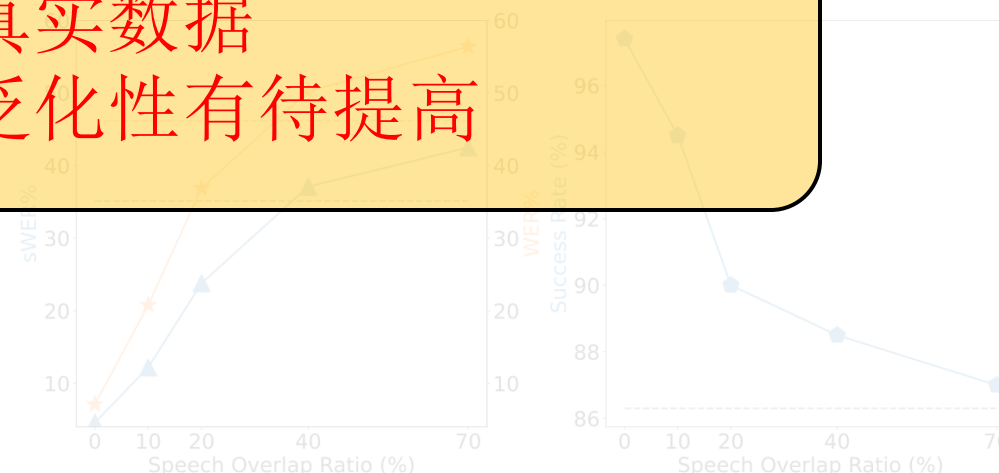
Method	$\Delta_a^\circ \downarrow$	$\Delta_e^\circ \downarrow$	$\Delta_d^\circ(\text{MAE}) \downarrow$
w/o IV	90.19	8.26	90.16
w/ IV ("Before")	1.60	1.76	2.70
w/ IV ("After")	3.35	2.56	4.69
w/ IV ("Before") & ST.	1.88	1.85	2.97

Spatial LibriSpeech [22]	-	-	6.50
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Method	Data	WER% $\downarrow$
w/o IV	LibriSpeech	8.6
w/o IV	Spatial LibriSpeech	9.0
w/ IV	Spatial LibriSpeech	7.6

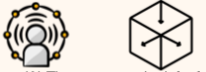
缺乏大规模真实数据
模型在真实场景的泛化性有待提高

Method	Data	SR% $\uparrow$	sWER% $\downarrow$	WER% $\downarrow$
w/o IV	left/right	50.1	4.4	67.2
	random	50.0	4.0	67.5
w/ IV	left/right	96.3	5.1	9.0
	random	80.2	5.4	29.8



引入空间信息，模型可以在SSL、FSR、LSE等任务上有不错的性能

MMAU-Pro

Spatial QA Question: Who's order does the waiter take first? Options: (A) The person to the left of the mic holder (B) The person to the right of the mic holder (C) The person in front of the mic holder  Answer: (A) The person to the left of the mic holder	Multicultural Music Question: What <i>raag</i> is this <i>bandish</i> composed in? Options: (A) Bhimpalasi (B) Kannada (C) Durga (D) Malkaunse (E) Bhairavi (F) Yaman Kalyan  Answer: (C) Durga	Voice QA Question: Answer the question in the audio. Options: (A) It sounds like this decision carries a lot of weight... (B) Perhaps the best approach is to systematically... (C) Your tranquil state suggests you have a high degree of mental clarity right now. This is an excellent time to trust your judgment, as it's likely unclouded by.... (D) Making major decisions during emotional peaks... Answer: Your tranquil state suggests you have a high degree of mental clarity right now. This is an excellent time to trust your judgment, as it's likely unclouded by emotional turmoil. 	Long Audio Question: What did the winning team get at the end of the game? Options: (A) They choose the next vacation destination (B) The other team has to get rid of their roster (C) They win the other team's apartment (D) The game ends in a tie and nothing changes Answer: They win the other team's apartment 		
Multi-Audio Question: What effect needs to be applied to the first recording to achieve sound of the second recording? Options: (A) echo (B) distortion (C) phaser (D) reverb  Answer: (D) reverb	Sound Question: What trend can be observed in the weight of the cloths thrown in the audio? Options: (A) Increasing (B) Decreasing (C) Remains constant (D) None of these options Answer: (A) Increasing  Perceptual Skills: Acoustic Trend Estimation Reasoning Skills: Temporal Reasoning	MMAU-PRO 		Voice STEM QA Audio Transcript: What is the packing efficiency, in percentage, of a solid where Atom X occupies the face-centered cubic lattice sites as well as alternate tetrahedral voids of the same lattice? Question: Choose the correct option that answers the question in the audio from the options given below. Options: (A) 25% (B) 35% (C) 55% (D) 75% Answer: (B) 35% 	Speech Question: In the audio with roughly the same phrase being repeated, explain how the different tone effects the meaning of each of the 4 phrases and in what order they occur. Options: (A) Bored → enthusiastic → questioning → angry (B) Cheerful → playful → serious → stern (C) Sarcastic → sincere → doubtful → irritated (D) genuine → sarcastic → questioning → frustrated. Answer: (D) He can remove 0 bones Perceptual Skills: Speech Activity, Turn-Taking and Overlap Detection Reasoning Skills: Quantitative Reasoning (Counting/Arithmetic Comparison) 
Multimodal Instruction Following Audio Transcript: Instruction: Explain what's happening in this audio. Your answer must contain a title, wrapped in double angular brackets, such as <<poem of joy>>. Question: What effect needs to be applied to the first recording to achieve sound of the second recording? Correct Answer: <<Harmonica's Ascending and Descending Scale>> This audio clip features a harmonica playing the C major scale. The musician first plays the scale in an ascending order, moving from the lowest note to the highest. Wrong Answer: This is an audio clip of a person playing the C major scale on a harmonica, first ascending and then descending. 	Music Question: Can you guess the singer in this song? Options: (A) Jeff Beck (B) Tenacious B (C) Jimmy Hendricks (D) Jack Black Answer: (D) Jack Black  Perceptual Skills: Timbre Perception and Instrument Recognition Reasoning Skills: Musicological Knowledge	Open-ended QA Question: What is hyper-foreignism with respect to pronunciation according to the clip?  Answer: When a speaker changes the way they say a word to sound more like that of the stereotype they hold for a foreign language	Speech-Sound-Music Mix Question: Which of the following songs is made according to the speaker? Options: (A) Not Like Us (B) Star Bly (C) All the stars (D) HUMBLE Answer: (D) HUMBLE 		

MMAU-Pro

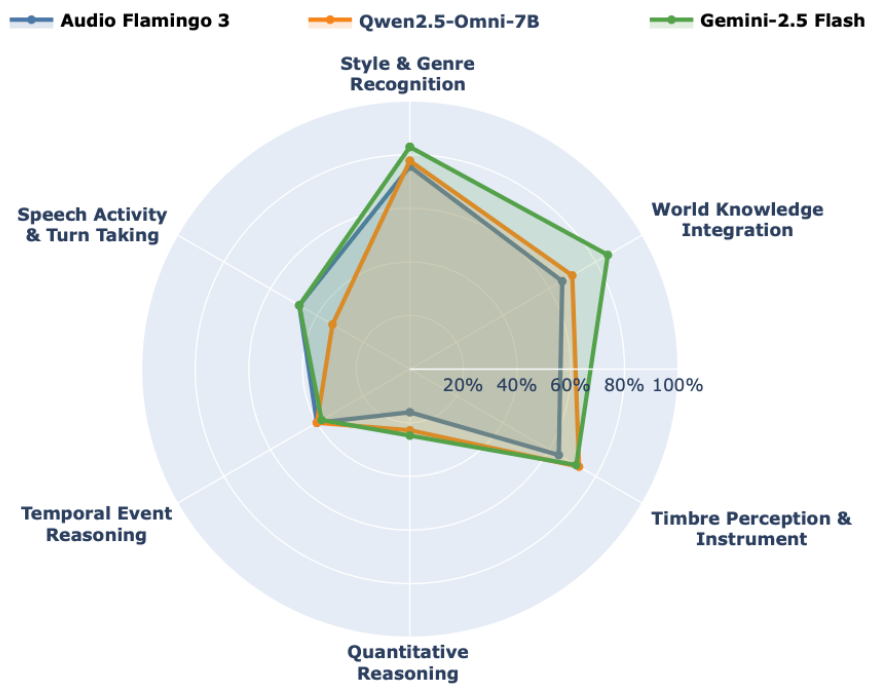
Capability	MMAU-Pro	MMAU	MMAR	AIR-Bench	AudioBench	MMSU	DynSuperb-1	DynSuperb-2
Long Audio Understanding	✓	×	×	×	✓	×	×	✓
Multi-Audio Understanding	✓	×	×	×	×	×	×	×
Spatial Audio Understanding	✓	×	×	×	×	×	×	✓
Open-Ended QA	✓	✓	✓	✓	✓	✓	×	✓
Multi-Step Reasoning	✓	✓	✓	×	×	✓	×	×
Multicultural Music	✓	×	×	×	×	×	×	×
Instruction Following	✓	×	×	✓	✓	×	✓	✓
In-the-wild Audios	✓	×	✓	×	×	×	×	✓
Voice Chat	✓	×	×	×	×	×	×	×
STEM Reasoning	✓	✓	✓	×	✓	×	×	×
Fully Human-Annotated	✓	✓	✓	×	×	✓	✓	✓

49中能力，新收集数据来源，包含长音频、多音频、空间音频理解，风格更丰富

MMAU-Pro

Models	Size	Sound	Music	Speech	Sound-Music	Speech-Music	Speech-Sound	Sound-Music-Speech	Spatial	Voice	Multi-Audio	Open-ended	IF	Avg.
Random Choice	-	28.3	26.1	29.4	24.2	25.2	30.5	14.8	21.2	29.3	25.2	-	-	23.4
Human	-	78.2	70.5	82.3	79.3	78.5	82.4	85.7	88.2	68.4	79.8	77.3	100	77.9
Large Audio Language Models														
SALMONN 7B	7B	32.2	44.9	38.3	22.0	34.8	28.4	28.6	26.5	36.5	11.4	31.2	33.9	34.5
SALMONN 13B	13B	43.6	47.2	37.3	28.0	47.8	38.4	42.8	30.8	53.2	17.4	33.6	38.5	39.6
GAMA	7.4B	45.4	41.2	29.8	24.0	27.9	27.3	14.8	12.0	28.4	20.2	24.2	31.7	33.2
DeSTA2	8.2B	31.0	43.3	46.5	32.6	47.8	39.7	42.8	32.6	54.8	13.2	25.4	41.5	36.7
DeSTA2.5-Audio	8.8B	35.7	48.2	49.9	22.0	36.9	35.2	28.6	28.0	51.0	19.8	36.4	46.5	40.6
BAT	7B	28.9	22.7	25.9	30.0	23.9	25.0	14.8	23.7	24.5	20.2	24.6	31.8	24.8
Audio Flamingo 2	3.2B	39.5	55.7	43.0	36.0	34.8	29.5	14.8	44.1	37.2	15.5	43.2	29.6	42.6
Phi4-MM	5.5B	25.7	47.8	47.6	30.0	39.1	30.1	28.6	39.7	42.7	11.4	42.5	65.4	38.7
Kimi-Audio	7B	46.0	57.6	52.2	46.0	54.3	48.9	42.8	43.7	50.6	17.2	34.5	42.3	46.6
Audio Flamingo 3	8.4B	55.9	61.7	58.8	40.0	41.3	47.7	57.1	26.8	58.6	26.0	44.2	33.3	51.7
Gemma-3n-E2B-it	5.1B	40.1	44.1	41.3	26.0	33.2	30.6	28.6	12.0	51.4	11.4	23.2	29.6	35.4
Gemma-3n-E4B-it	7.5B	42.4	46.4	44.9	38.0	45.6	31.8	57.1	21.8	58.3	19.6	28.5	36.4	39.7
GPT4o-mini-Audio	-	40.2	59.7	66.1	35.3	42.2	55.9	42.8	12.0	52.7	22.4	41.6	79.7	48.3
GPT4o-Audio	-	44.7	63.1	68.2	40.4	43.5	62.5	57.1	21.4	57.5	32.6	43.2	82.5	52.5
Large Audio Reasoning Models														
R1-AQA	8.2B	47.9	31.9	33.7	32.0	36.9	20.4	28.5	23.6	32.7	11.4	38.5	44.2	34.1
Audio-Reasoner	8.4B	34.2	50.1	44.0	26.0	36.9	43.2	28.6	20.3	43.4	22.6	38.6	43.4	39.5
Mellow	167M	27.6	32.9	27.9	24.0	34.8	27.3	14.3	23.7	28.3	20.8	21.4	23.5	27.5
Omni Models														
Ming-Lite-Omni-1.5	18.9B	47.9	56.2	49.1	30.0	39.1	45.4	42.8	31.7	44.5	37.4	42.7	48.2	47.4
Baichuan-Omni-1.5	7B	34.6	32.5	36.5	30.0	19.5	30.7	28.5	21.2	40.0	28.8	39.7	47.2	33.9
Qwen2.5-Omni-3B	5.5B	38.5	60.3	53.9	40.0	45.6	46.6	42.8	28.9	46.5	11.4	47.6	58.4	46.1
Qwen2.5-Omni-7B	10.7B	47.6	61.5	57.4	40.0	53.2	60.2	28.5	41.2	60.0	24.3	52.3	61.3	52.2
Gemini-2.0 Flash	-	48.4	56.9	69.5	39.6	57.6	55.9	42.8	34.6	68.6	26.5	66.8	94.2	55.7
Gemini-2.5 Flash	-	51.9	64.9	73.4	42.8	58.7	61.3	42.8	36.3	71.7	21.2	67.5	95.1	59.2
Cascaded Systems														
Caption + GPT4o	-	38.6	40.6	38.4	21.6	38.2	25.5	28.6	9.5	38.6	24.7	27.6	88.2	35.3
Captions + Qwen235B-A22B	235B	36.4	41.3	36.1	18.6	37.4	24.5	14.3	5.8	35.6	22.5	25.6	85.5	33.7

MMAU-Pro



- 音乐风格类型
- 知识性问题
- 音色乐器类型



- 语音、音频动态
- 时序因果推断
- 定量问题

通用音频感知大模型面临的挑战

- 如何处理好音频和文本模态的耦合关系，使模型不降智仍未解决
- 更广覆盖的音频数据、高质量的空间音频数据仍然有限
- 多段音频理解推理有待提升
- 模型推理能力的探索仍有空间（基于音频的推理）

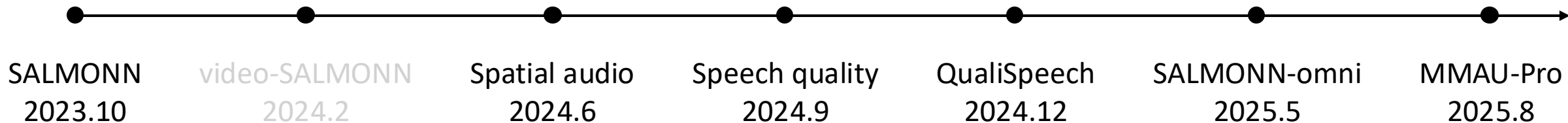
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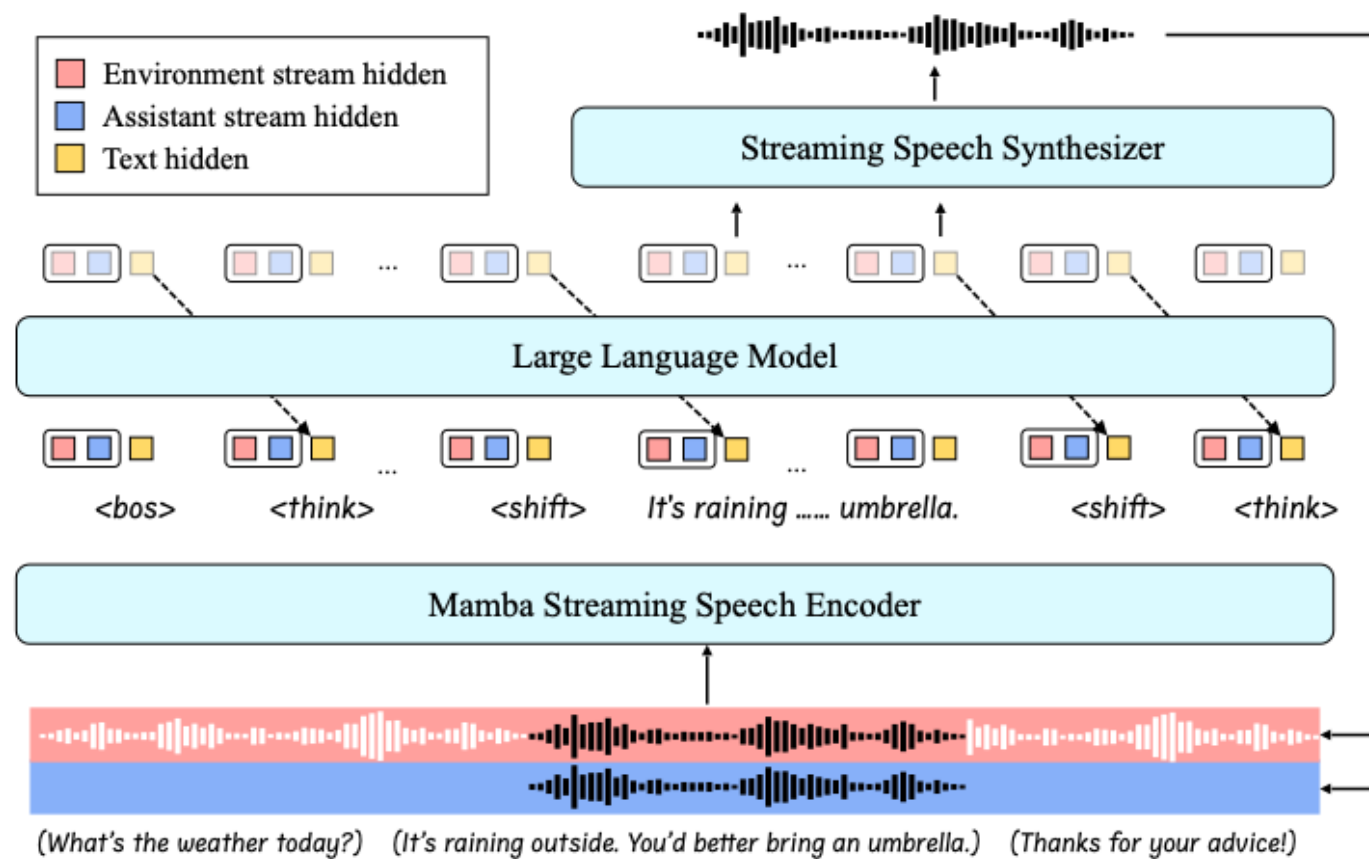


无codec注入的全双工对话框架

实现全双工语音对话模型面临4个难点：

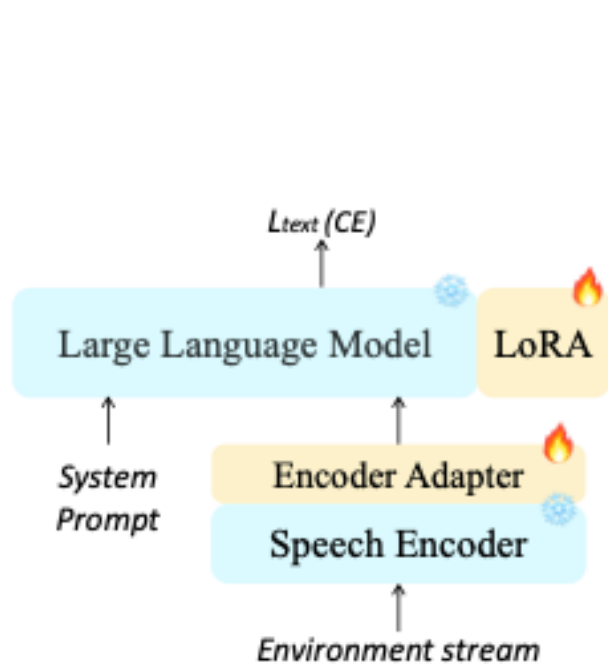
- 支持流式语音输入和输出
引入流式音频编码模块和生成模块
- 能同时处理环境流和模型回复流
将音频流和文本流交错成一个序列
- 引入时间的概念对齐音频和文本模态
将输入输出划分成一系列对应定长时间的“时间块”
- 能处理自然对话中的复杂现象（如barge-in, backchanneling）
引出“思考”机制

SALMONN-omni

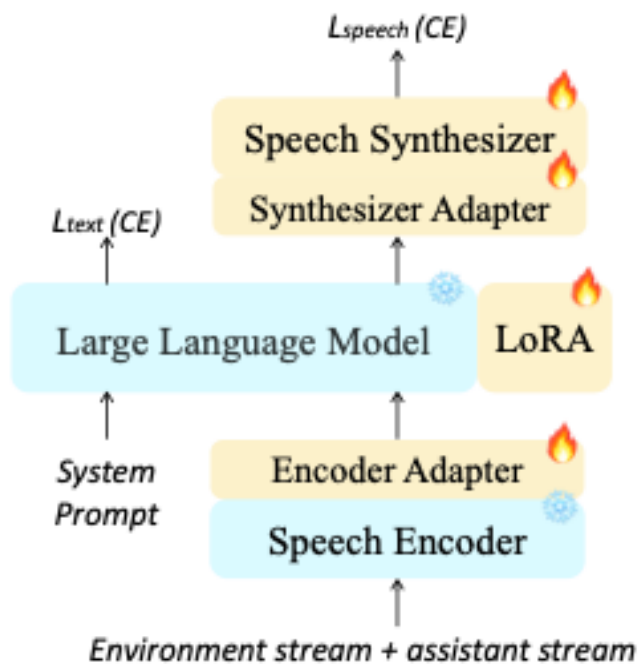


Yu et al., A Standalone Speech LLM without Codec Injection for Full-duplex Conversation, NeurIPS 2025.

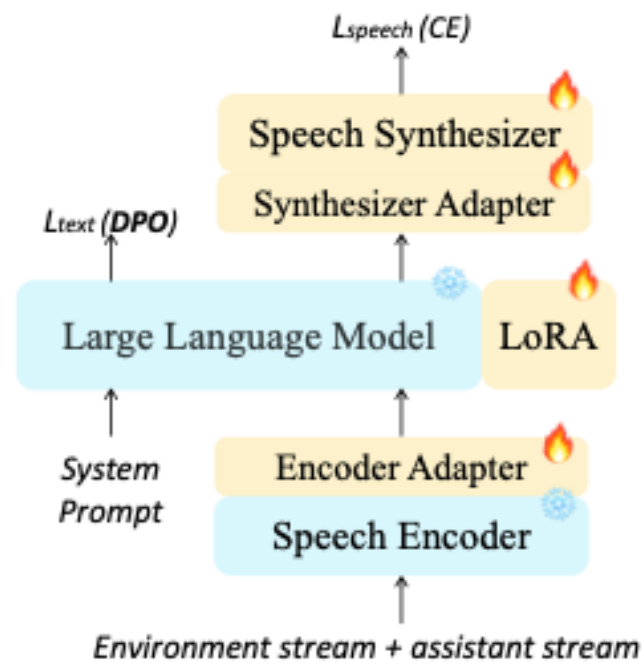
SALMONN-omni



(a) Stage1
Connecting Encoder

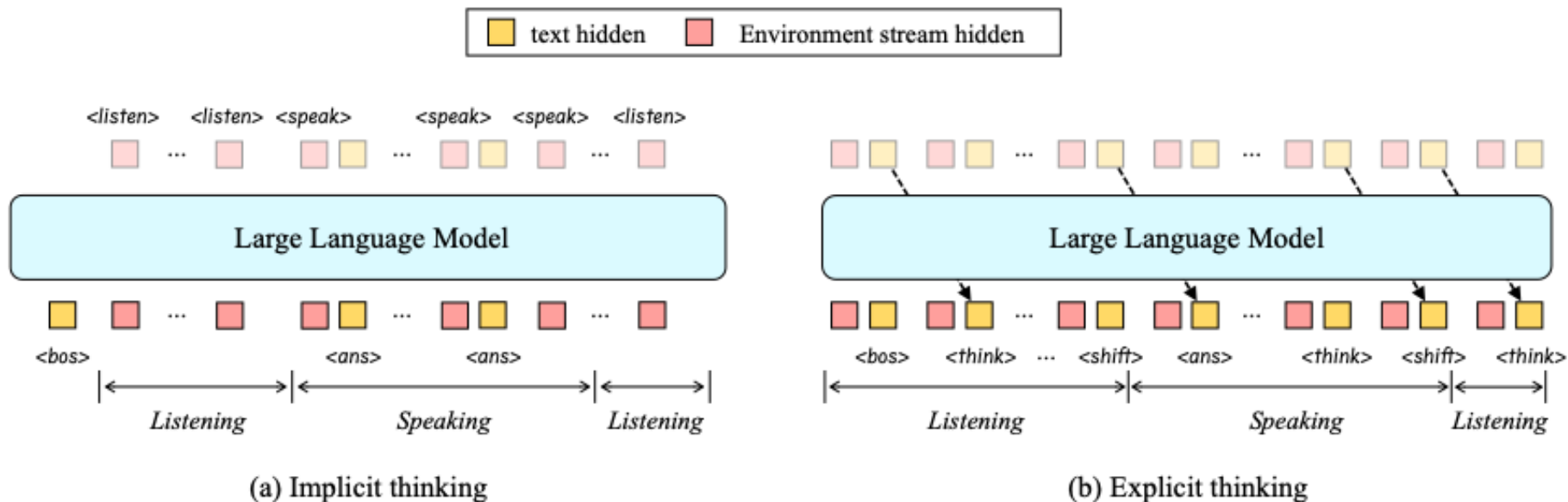


(b) Stage2
Connecting Synthesizer



(c) Stage3
RL for full-duplex modeling

隐式 v.s. 显式思考



Full-duplex strategy	Llama Q.		Web Q.		TriviaQA		AlpacaEval	
	succ. %	S2T	succ. %	S2T	succ. %	S2T	succ. %	S2T
Implicit	97.3	76.3	99.9	50.8	95.8	62.0	67.3	3.73
Explicit	100.0	82.0	99.9	52.4	92.3	67.5	68.3	4.48

语音问答性能

Model	Llama Q.		Web Q.		TriviaQA		AlpacaEval		Avg. UTMOS
	S2T	S2S	S2T	S2S	S2T	S2S	S2T	S2S	
Predicted turn-taking									
Moshi [11]	60.8	54.5	23.4	22.1	25.6	16.7	1.84	1.76	3.178
Freeze-Omni [20]	74.2	56.2	40.8	27.9	45.1	28.5	3.90	2.46	4.277
SALMONN-omni	79.3	73.6	49.7	43.7	63.6	56.0	4.01	3.22	4.255
Oracle turn-taking									
GLM-4-Voice [15]	75.0	65.7	38.5	37.0	50.8	47.5	3.82	3.58	4.081
Qwen2.5-Omni [22]	78.7	75.7	41.9	39.6	55.5	52.2	3.04	2.61	4.085
VITA-1.5 [19]	77.3	56.0	43.9	31.0	54.8	41.3	3.92	1.94	4.022
miniCPM-o [26]	76.3	70.0	47.1	42.7	65.4	56.0	3.99	3.44	4.147
Kimi-Audio [21]	79.7	68.3	44.0	37.3	63.6	51.2	3.92	2.95	2.679
Baichuan-Audio [27]	76.0	74.0	43.1	40.7	56.9	53.0	3.76	3.32	4.182
Freeze-Omni [20]	77.0	61.0	42.3	29.1	53.5	36.5	3.71	2.42	4.276
SALMONN-omni	80.0	75.0	50.5	45.4	66.0	58.8	4.05	3.33	4.261

双工能力测试

Model	Llama Q.	Web Q.	TriviaQA	AlpacaEval
Moshi	85.0	76.0	37.1	83.4
Freeze-Omni	99.7	99.8	72.0	87.9
SALMONN-omni	99.7	99.6	92.8	92.0

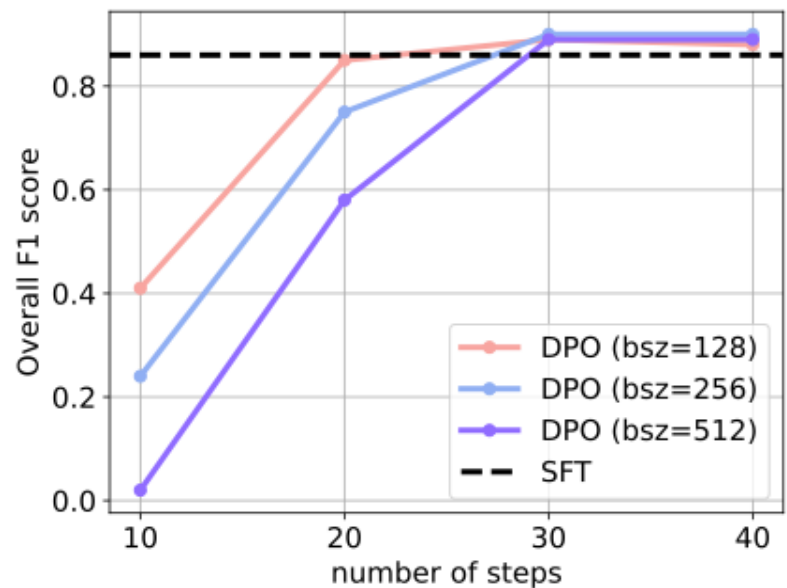
Turn-taking预测

Model	Echo factor	Precision	Recall	F1 score
Moshi	$\times 1.0$	0.84	0.77	0.80
Freeze-Omni	$\times 0.0$	0.64	0.72	0.68
Freeze-Omni	$\times 0.1$	0.19	0.86	0.31
Freeze-Omni	$\times 1.0$	0.10	0.69	0.17
SALMONN-omni	$\times 1.0$	0.84	0.92	0.88

上下文无关的打断
backchanneling

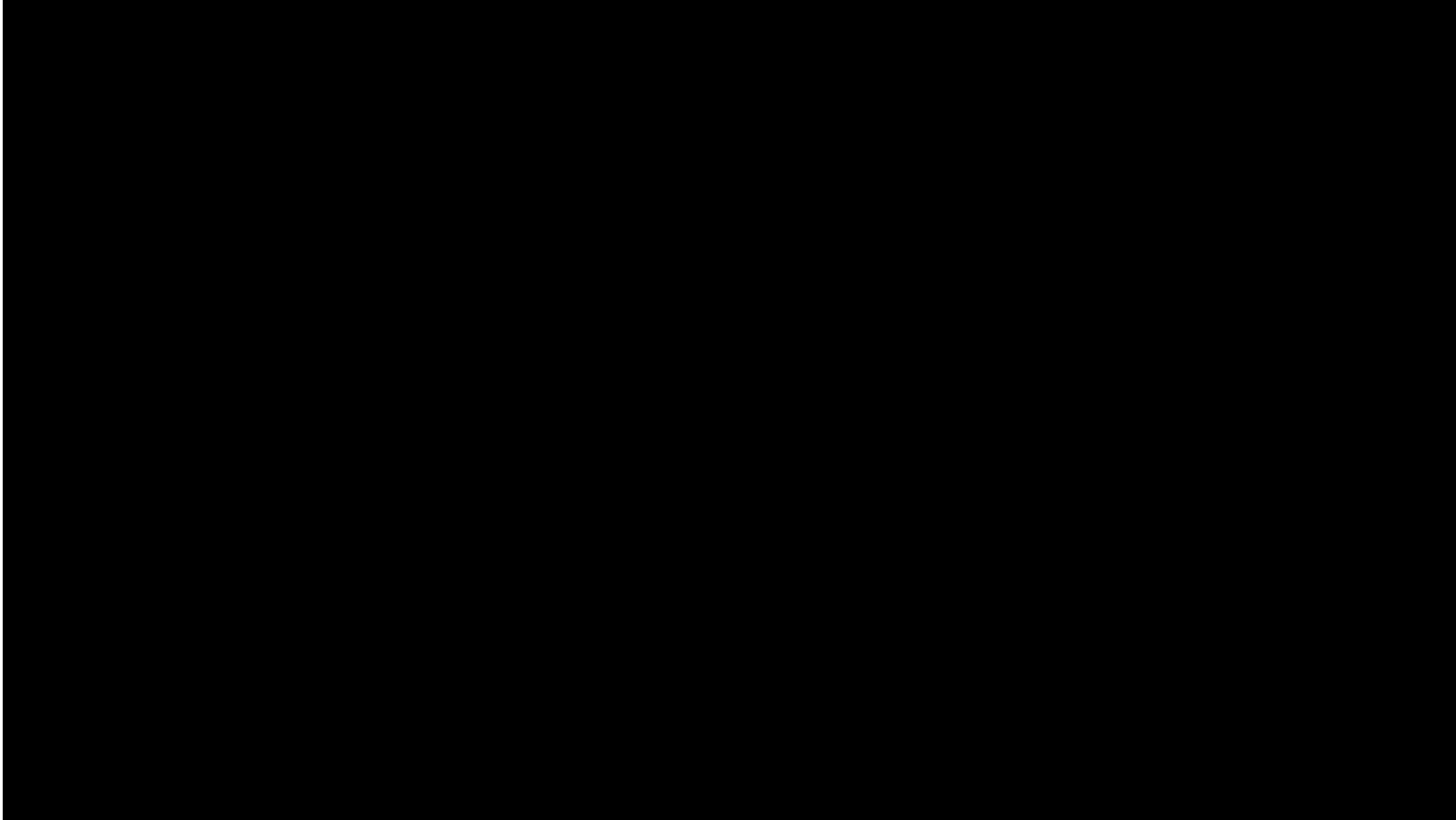
双工能力测试

Stage	Contextual-Independent			Contextual-Dependent			Overall F1 score
	Precision	Recall	F1 score	Precision	Recall	F1 score	
SFT (Stage 2)	0.68	0.93	0.79	0.88	0.99	0.93	0.86
DPO - 10 steps	0.88	0.07	0.13	0.95	0.21	0.34	0.24
DPO - 40 steps	0.84	0.92	0.88	0.89	0.98	0.93	0.90

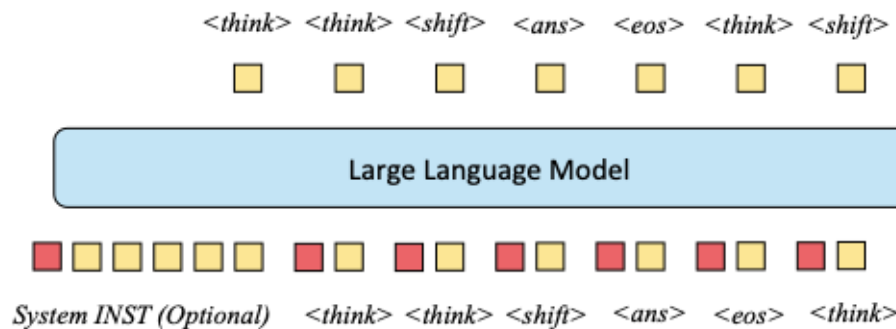


使用DPO削弱模型对被打断的偏好，
增强对上下文的理解能力

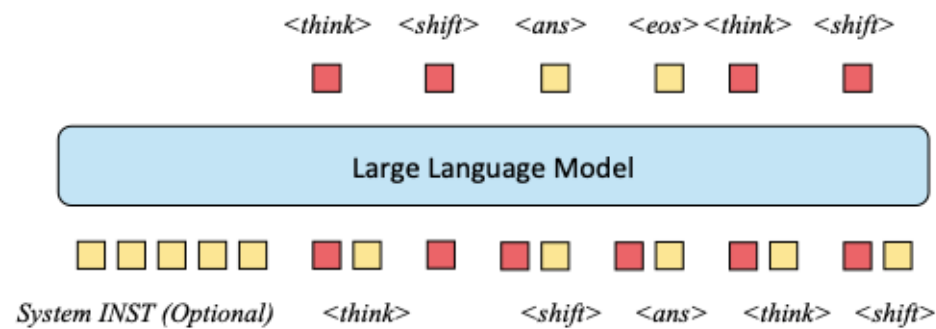
SALMONN-omni



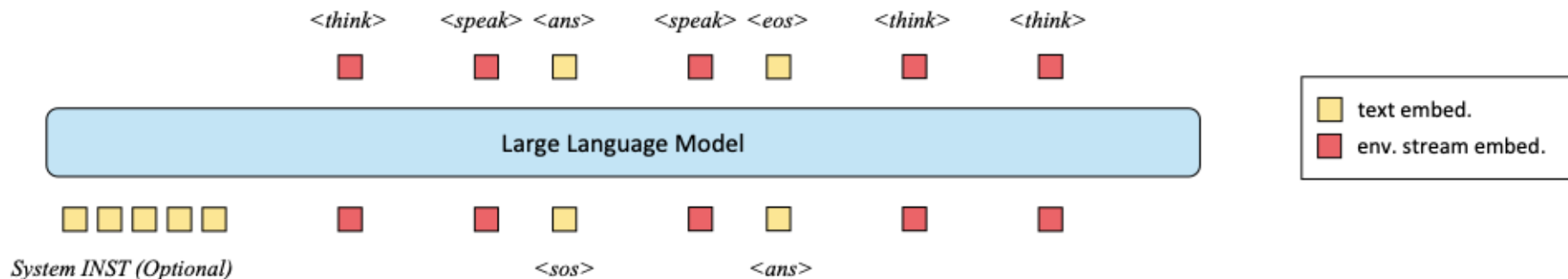
思考内容



(a) Explicit text-driven thinking



(b) Explicit speech-driven thinking



(c) Implicit speech-driven thinking

■ text embed.
■ env. stream embed.

思考内容

- ASR

.....*He.....is.....the.....one.....with...<shift>He is the one with the worst record*

- Phoneme

.....*h I h I z ɔ̃ ə w ʊ n w I ɔ̃<shift>He is the one with the worst record*

- N-Best

.....*but but which which but.....which which had had which.....had had been been had.....been been the the been.....the the occasion occasion the<shift>but which had been the occasion of the enigmatic scene I had witnessed at the...*

思考内容

“Thinking” Strategy	test-clean		test-other		Llama Questions		Web Questions	
	%WER (↓)	%SR	%WER (↓)	%SR	%Acc. (↑)	%SR	%Acc. (↑)	%SR
Implicit-Speech-SP	2.67	93.3	6.63	94.6	80.0	55.7	50.6	89.0
Implicit-Speech-ASR	3.48	94.7	8.26	95.7	79.7	77.7	46.8	98.6
Explicit-Speech-SP	2.57	95.5	6.48	96.5	83.0	85.0	48.4	98.5
Explicit-Text-SP	2.40	95.6	6.09	95.6	83.3	87.7	52.7	99.9
Explicit-Text-ASR	5.09	91.8	9.85	94.2	72.3	64.0	40.3	98.5
Explicit-Text-NS	7.65	91.8	10.5	92.9	74.3	86.0	35.6	99.9

“Thinking” Strategy	Precision	Recall	F1 score
Implicit-Speech-SP	—	—	—
Implicit-Speech-ASR	0.46	0.93	0.61
Explicit-Speech-SP	0.78	0.93	0.85
Explicit-Text-SP	0.76	1.00	0.87
Explicit-Text-ASR	0.76	0.90	0.82
Explicit-Text-NS	0.59	1.00	0.74

“Thinking” Strategy	test-clean	test-other	%SR
Explicit-Text-ASR	4.52	9.26	94.7
Explicit-Text-PH	3.94	8.86	95.1
Explicit-Text-NB	3.65	8.15	95.6

思考内容

Describe... okay, this is a general question, not too specific, but I'll give it a shot. What are dreams? That's a pretty broad topic... \n\n pause *\n\n...are. Okay, so it's a question asking me to describe something, not just define it. That's a bit more challenging. I need to think about what dreams are, how they work, what they're for...<shift>The mysterious world of dreams! Dreams are a universal human experience, and yet, they remain somewhat of an enigma. Here's a description of what dreams are... ..*

如何让模型思考的内容有意义、有帮助，且不会显著
增加延迟仍待解决

model	Llama Questions
baseline	65.0 (31.7)
DPO – duplex	66.7 (74.4)
DPO – duplex + correctness	75.3 (17.7)

语音合成模块

- 纯codec-based模型性能更易达到上限
- 纯flow-matching模型难以微调到与LLM对齐
- 自回归+扩散模型性能较好，但Time-To-First-Audio (TTFA)较高
- 双工模型中更需要较低的TTFA，而非RTF

Thanks !